

HAT-P-18b

R-2513

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_170627 · created 2026-08-25T00:28:47+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457931.71228325 +/- 6.5e-07 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1836466 +/- 1.4e-06
Transit depth (Rp/Rs)^2	3.37 +/- 5.3e-05 [%]
Orbital Inclination (inc)	87.2077 +/- 0.0018
Airmass coefficient 1 (a1)	0.035 +/- 0.02
Airmass coefficient 2 (a2)	2.9999996 +/- 4e-06
Scatter in the residuals of the lightcurve fit is	48.16 %
Best Comparison Star	#3 - [527, 73]
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.07 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1836466 $\pm$ 1.4e-06 vs 0.1356 (deviation 54.51 σ)
Duration fit vs published	0.07 d vs 0.1129 d (deviation inf σ)
Mid-time O-C	0.3 min
Depth SNR	208.3
Residual scatter	48.16 %

Light curve

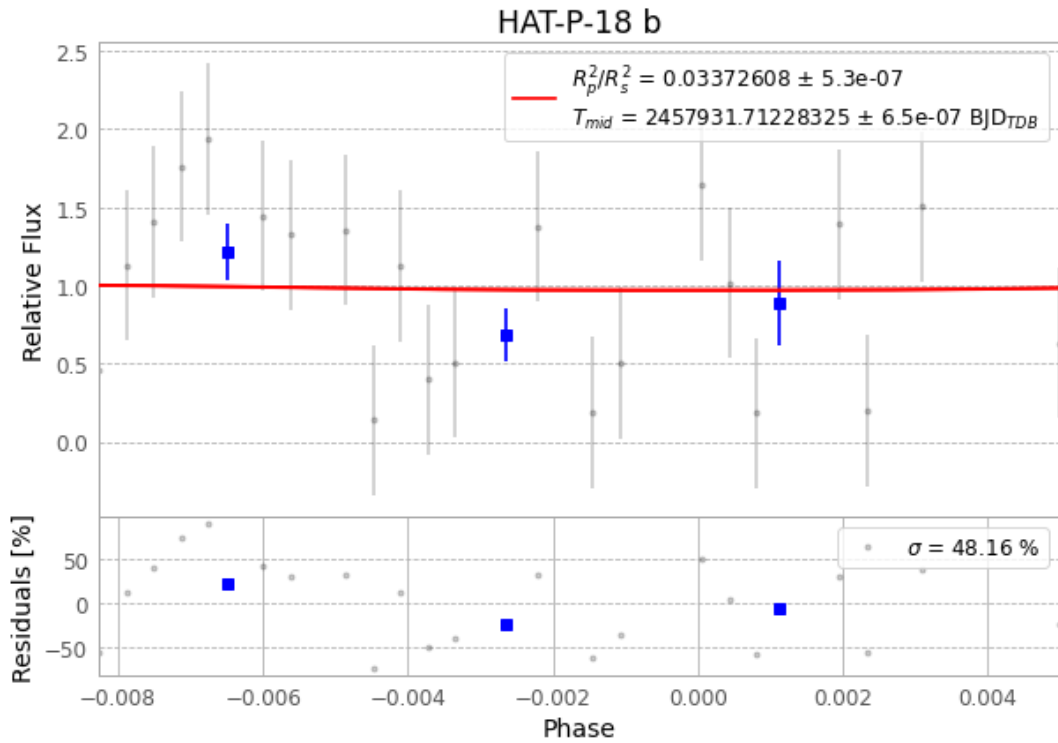


Figure 1 — FinalLightCurve_HAT-P-18 b_2017-06-26.png (SHA-256 c45126eda9dc3523...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [243, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 65fa31ff20f532e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

42 FITS frames (28 MB), HATP-18170627035412.FITS → HATP-18170627055713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-170627.log (878955940ab3...), FinalLightCurve_HAT-P-18 b_2017-06-26.png (c45126eda9dc...), FinalLightCurve_HAT-P-18 b_2017-06-26.csv (e69c90aaf017...)

Compute resources

Wall time 148.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)